

Lecture #16

Date: Oct 14, 25

Day: Tuesday

Minimum currency change.

\$1	\$5	\$2	\$3
-----	-----	-----	-----

\$1 coin exist
in the array

∴ Place it on 1st index
for base case

Amount $\Rightarrow$ \$14

if we have multiple occurrences of the coins

(4, \$14)

6 drop

use

(3, \$14)

(4, \$11)

include

(2, \$14) $\xrightarrow{5} +1$

(1, \$14)

(2, \$9)

(4, \$9) $\xrightarrow{4+1} 9$ min

(1, \$9)

(2, \$4)

4

(1, \$4)

4

(3, \$8)

(2, \$8) $\xrightarrow{2+1} 3$

4,
(4, \$8)

(4, \$6)

(3, \$6)

(4, \$6)

\$1	\$3	\$5	\$10	\$2
-----	-----	-----	------	-----

CC (N, Amt) { // Brute Force Recursive

if (N == 1) // index = \$1
return Amt

if (Amt == 0) // Amt == 0
return 0

if (C[N] > Amt) // Coin Drop
return CC(N-1, Amt)

else

return min(CC(N-1, Amt), CC(N, Amt - C[N]) + 1)

} $T(N) = (2^N * 2^{Amt})$

Bottom up Approach :

Res[0....N][0....Amt]

for j = 0 to Amt
Res[1][j] = j

for i = 0 to N
Res[i][0] = 0

for $i = 2$ to N

for $j = 1$ to Amt

if $(C[i] > j)$

$R[i][j] = R[i-1][j]$

else

$R[i][j] = \min(R[i-1][j], R[i][j-C[i]] + 1)$

return $R[N][M]$;

$T = O(N * Amt)$
 $= O(N * M)$

0 1 2 Amt 3 4 5

Lecture #17

Date: 21 Oct 2025

Day: Tuesday

Fractional Knapsack (Greedy Algos)

$$M = 10$$

val (\$)	1000	500	250	180
wt (kg)	8	5	4	3

Greedy choice: pick max value item first

* sort the value array in descending order, so the max. value is at start

* Greedy solutions r not always optimal.

Counter example (where greedy choice fails)

value (\$)	{ 1000, 500, 300, 200, 100 }
wt (kg)	{ 10, 3, 4, 2, 1 }

greedy choice: \$1000 (max. first)

but max. profit is: $3 + 4 + 2 + 1$

= \$1100.

so this greedy sol doesn't give max. profit

val (\$)	= { 50, 80, 100, 200, 300, 800 }
wt (kg)	= { 2, 3, 4, 5, 6, 8, 10 }

Greedy choice: pick min. weight first.

$$\rightarrow 50 + 80 + 100 + \left(\frac{1}{5} \times 200\right) = 270 \$$$

not maximum: so this greedy choice also fails

for optimal solution:

Greedy choice: compute per unit value,
pick maximum of that first.

value(\$): { 20, 50, 80, 100, 200, 300, 800 }
wt (kg): { 2, 3, 4, 5, 6, 8, 10 }

$v/w = \{10, 16.6, 20, 20, 33.3, 37.5, 80\}$

fn knsp (wt[1...N], val[1...N], M, N) {

profit = 0

$v-w[1 \dots N]$

for ($i = 1$ to N) — $O(N)$

$v-w[i] = v[i] / w[i]$

sort ($v-w, N, val, wt$) — $N \log N$

// sort on base (v/w) ratio in desc order

for ($i = 1$ to N) { — $O(N)$

if ($M \geq wt[i]$)

$M = M - wt[i]$

profit += val[i]

else

profit = profit + $\left(\frac{M}{wt[i]} \times val[i] \right)$

break ;

}
return profit

Date: Oct 23, 25

Day: Thursday

Activity Problem

start-time: 1, 3, 0, 5, 3, 5, 6, 7, 8, 2, 12

finish-time: 4, 5, 6, 7, 9, 9, 10, 11, 12, 14, 16 → sorted

* Greedy choice: earliest finish time.

Finish time earlier → scheduled first

count = 1

k = 1 // (index of last selected activity)

for i = 2 to N

if (s-time[i] ≥ f-time[k])

k = i

count++

11
8
4
1

return count.

* Greedy choice: maximum start time first
// sort the -time in desc. order

s-time: 12, 8, 7, 6, 5, 5, 3, 3, 2, 1, 0

f-time: 16, 12, 11, 10, 7, 9, 5, 9, 14, 4, 6

(12-16) (8-12) (5-7) (3-5)
4 activities

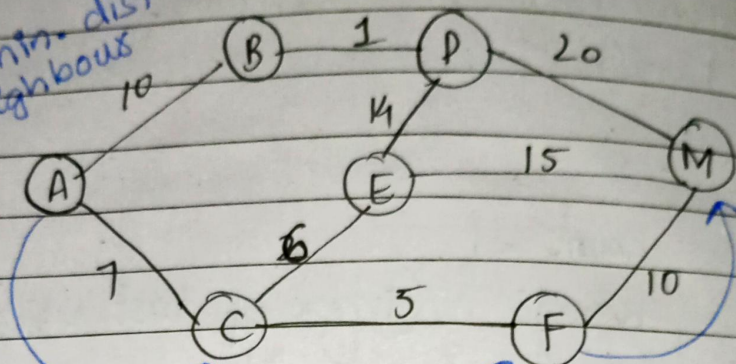
Both choices are optimal.

When Greedy choice is optimal?

1. Optimal substructure
2. Local optima provides global optima.

A → M
if we pick min. dist.
to the neighbours

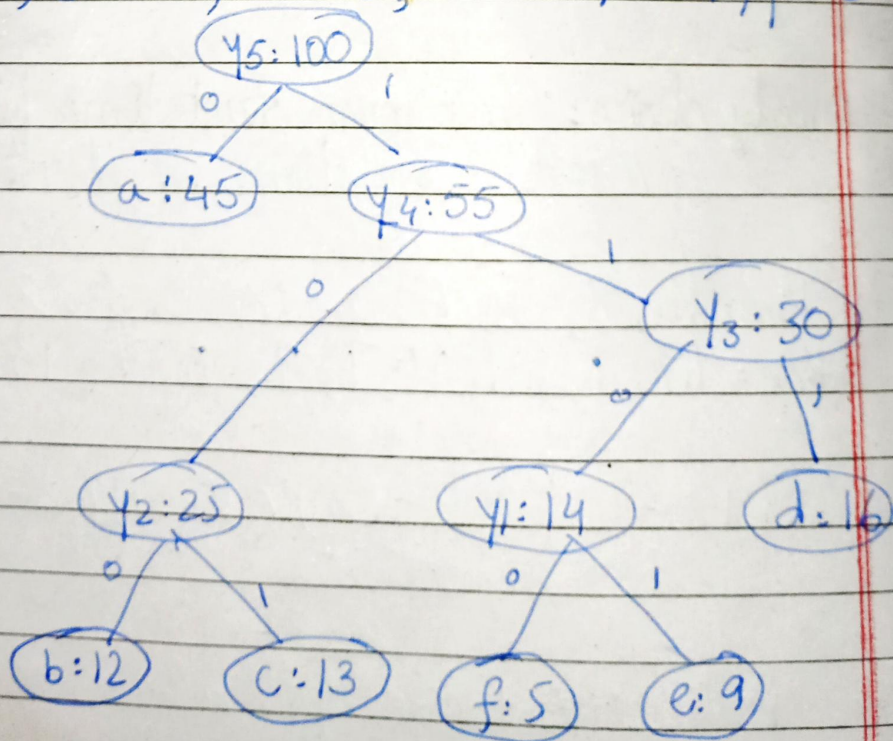
counter
example



if we had $C \rightarrow E$ distance
2 then it will be optimal

Huffman encoding:-

$a=45$, $b=12$, $c=13$, $d=16$, $e=9$, $f=5$



Date: _____

Graphs:

Vertices

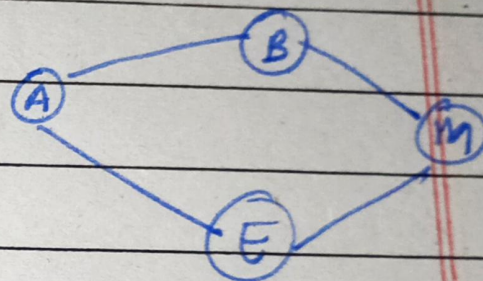
Edges

	A	B	C	D	E	F	M
A	0	10	7	-	-	-	-
B							
C							
D							
E							
F							
M							

unweighted:

	A	B	E	M
A	0	1	1	0
B	1	0	0	1
E	1	0	0	1
M	0	1	1	0

V2



Directed Graph:

